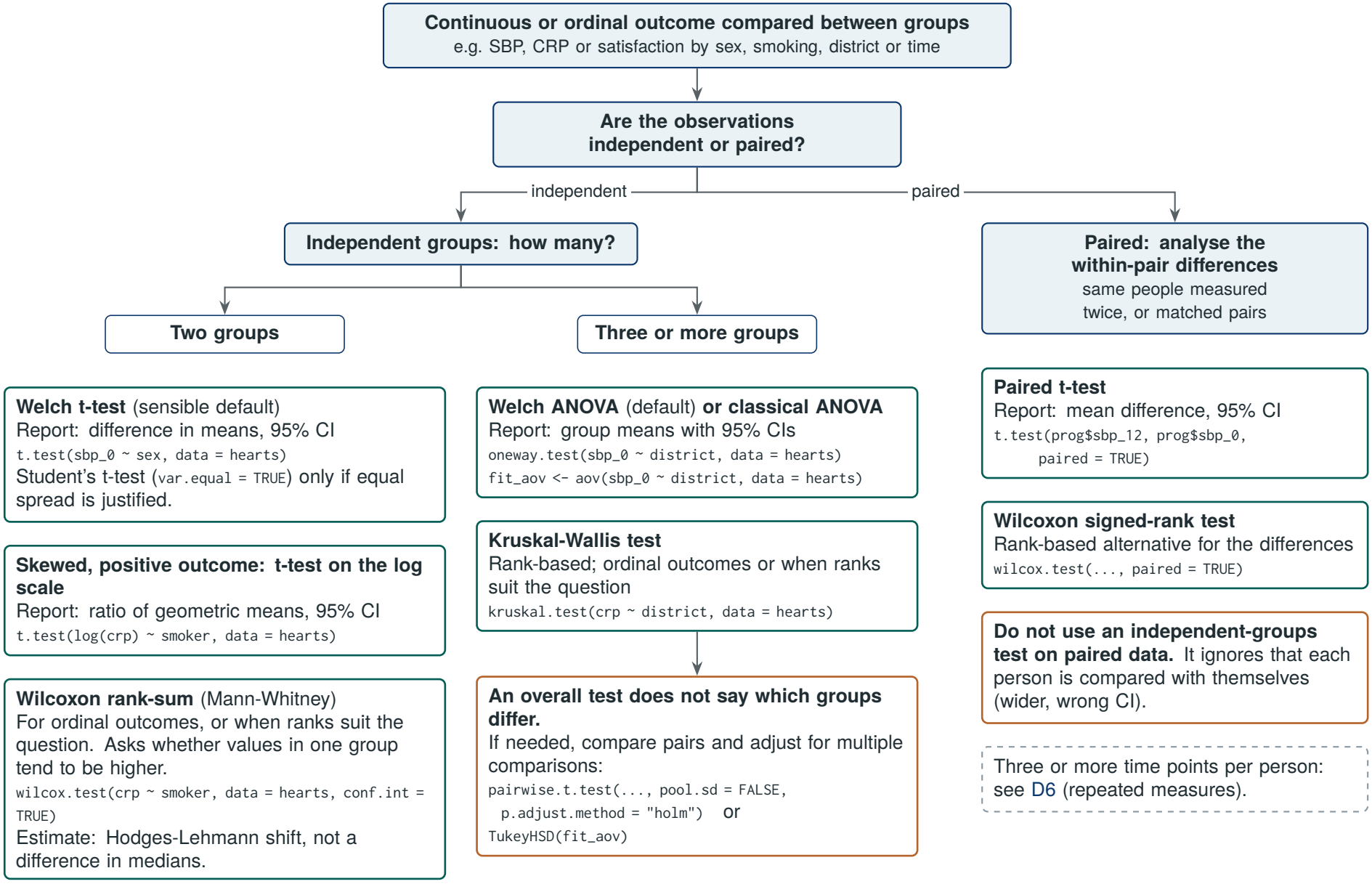
D2. Comparing a continuous or ordinal outcome between groups

Report the size of the difference with a 95% CI. The test is only one part of the answer.



Assumptions and choices. t-tests and ANOVA assume independent observations (from the design) and approximately normal sample means. In large groups this holds even with moderate skew; with small groups, strong skew or outliers matter more. Check residual plots. Rank-based tests are not assumption-free: they also need independent observations, and they are not in general tests of medians. Do not choose a test from sample size alone or from a normality-test p-value: look at the data and decide which summary answers your question.